

Charge Densities in Real Space: Primitive Ontology and the Individuation of Electrons

Claes Johnson¹

¹KTH Royal Institute of Technology, SE-100 44 Stockholm, Sweden ,
claesjohnson@gmail.com

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Abstract

We compare the ontology of an atomic system of N electrons under two models. Standard quantum mechanics treats the electrons as *identical particles* and represents the system by a probability amplitude over their configurations in a $3N$ -dimensional configuration space, anti-symmetric under exchange. *RealQM* represents it by N unit charge densities, each carried on its own region Ω_i of ordinary three-dimensional space, the regions non-overlapping ($\Omega_i \cap \Omega_j = \emptyset$). Two long-standing metaphysical problems — the *arena* of the wave function and the *indivisibility* of identical particles — attach to the first model, one to each of its two features: the $3N$ -dimensional domain and the antisymmetry. The second model has neither feature and so incurs neither problem. It is a primitive ontology in three-space (no configuration-space object exists), and it *individuates* its electrons absolutely, by the regions they occupy (the Principle of the Identity of Indiscernibles is satisfied); the two virtues are one fact, the non-overlap of the Ω_i doing double duty. RealQM is moreover *constructive*: its charge densities are the objects one actually computes, applied to real chemistry and nuclear structure. We state both ontologies in equations, place RealQM against the Standard Model's point-and-field ontology for perspective, and bound the claim: RealQM is a rival theory, empirically warranted in its atomic and nuclear domain, and the metaphysical verdict is conditional on that adequacy.

1 Two problems in one formula

An atomic system of N electrons admits two descriptions. Standard quantum mechanics treats the electrons as *identical particles* and represents the system by one probability amplitude over their joint configurations, in a space of $3N$ dimensions. RealQM treats them as N *individual charge densities*, each carried on its own region $\Omega_i \subset \mathbb{R}^3$, in ordinary three-dimensional space (§2). This contrast — identical particles in a high-dimensional configuration space against individuated densities in ordinary space — is the whole of what follows. We take the standard side first.

Write its state:

$$\Psi : \mathbb{R}^{3N} \rightarrow \mathbb{C}, \quad \Psi(\dots, \mathbf{x}_i, \dots, \mathbf{x}_j, \dots) = -\Psi(\dots, \mathbf{x}_j, \dots, \mathbf{x}_i, \dots). \quad (1)$$

The domain is $\mathbb{R}^{3N}$; the minus sign is the *antisymmetry* of the amplitude under exchange of any two particles — Fermi–Dirac statistics, the Pauli principle in its formal guise. Two metaphysical difficulties follow directly, one from each of these two features of (1).

The arena, from the domain $\mathbb{R}^{3N}$. If Ψ is the fundamental object — wave-function realism [1, 2, 3] — then ordinary three-space is not fundamental but must be recovered from a field on a $3N$ -dimensional space. The rival *primitive ontology* programme [4, 5, 6] insists instead on beables in $\mathbb{R}^3$, with Ψ demoted to a dynamical or nomological role. Either way the $3N$ -dimensional domain is the source of the difficulty.

Individuality, from the antisymmetry. The permutation symmetry in (1) makes the particles, on the received reading, *non-individuals*: no fact settles which is which, and two electrons in one atom are numerically distinct yet share all properties, in apparent violation of Leibniz's Principle of the Identity of Indiscernibles (PII) [7, 8]. Defenders of PII retreat to *weak discernibility* by irreflexive relations, with no distinguishing monadic property [9, 10].

Both problems are consequences of the two choices in (1): the domain $\mathbb{R}^{3N}$, and the antisymmetry. A formulation making neither choice does not incur them.

2 A real-space ontology, in equations

Take instead the RealQM formulation [11]

$$\Psi(\mathbf{x}) = \sum_{i=1}^N \psi_i(\mathbf{x}), \quad \mathbf{x} \in \mathbb{R}^3, \quad \text{supp } \psi_i = \overline{\Omega_i}, \quad \Omega_i \cap \Omega_j = \emptyset \ (i \neq j), \quad (2)$$

where ψ_i^2 is the unit charge density of electron i on its own region $\Omega_i \subset \mathbb{R}^3$ and $\rho = \sum_i \psi_i^2$ is the total electron density. For a nucleus (kernel) of charge Z at the origin, with attractive potential $V(\mathbf{x}) = Z/|\mathbf{x}|$, the ground state minimises the energy

$$E = \sum_i \frac{1}{2} \int_{\Omega_i} |\nabla \psi_i|^2 - \int V \rho \, dx + \frac{1}{2} \iint \frac{\rho(x)\rho(y)}{|x-y|} \, dx \, dy, \quad \int_{\Omega_i} \psi_i^2 \, dx = 1, \quad (3)$$

over the ψ_i and the free boundaries $\partial\Omega_i$.

RealQM is, at once, a *primitive ontology in three-space*: what exists is the extended charge density ρ in $\mathbb{R}^3$, and there is *no* object on $\mathbb{R}^{3N}$ to interpret or to recover three-space from — (1)'s domain is simply absent. Unlike primitive-ontology theories that carry a matter density *alongside* a configuration-space Ψ [5], here the density is the whole matter ontology and (3) its law. It is a rival theory, not a reinterpretation of (1), warranted by the chemistry and nuclear structure it computes [11].

3 Individuation by occupation

The basic objects of (2) are extended and disjoint, and this individuates them. Define electron i as the density on Ω_i ; then occupancy of Ω_i is a *monadic* property that i has and $j \neq i$ lacks, since $\Omega_i \cap \Omega_j = \emptyset$. Hence

$$i \neq j \iff \Omega_i \neq \Omega_j : \quad \text{no two electrons share all properties.} \quad (4)$$

The Principle of the Identity of Indiscernibles, standardly taken to fail for (1), is *satisfied*. Moreover the electrons are *absolutely* discernible — each has a distinguishing monadic property (its region)

— which is stronger than the *weak* discernibility [10] to which PII’s defenders retreat under (1). Individuality in the full classical sense is recovered.

And extension and individuation are *the same fact*. The disjointness $\Omega_i \cap \Omega_j = \emptyset$ that keeps each ψ_i a density in $\mathbb{R}^3$ (the primitive-ontology virtue) is the very disjointness that discerns the electrons (the individuality virtue). Where (1) encodes exclusion as antisymmetry — a fact about a configuration-space function — (2) encodes it as impenetrability — a fact about *space*: two charge clouds cannot occupy one place. Pauli exclusion and the identity of indiscernibles, distinct in the received picture, are one geometric fact here.

4 Computability: the ontology is the method

The standard amplitude (1) lives on $\mathbb{R}^{3N}$, and representing it there costs exponentially in the number of particles: solving (1) directly is feasible only for very small N , and beyond a few electrons one must retreat to approximations (Hartree–Fock, density-functional theory). RealQM (2)–(3) lives instead on ordinary $\mathbb{R}^3$, where complexity scales with mesh points in three dimensions, not with the exponential dimension of $\mathbb{R}^{3N}$; it is directly computable for real systems.

This also sets RealQM apart from the metaphysical proposals of §1. Wave-function realism and the primitive-ontology programme are readings of a formalism one solves — approximately — on $\mathbb{R}^{3N}$; the ontology is interpretive, supplied after the calculation. In (2)–(3) the beables *are* the objects of computation: one minimises the energy (3) directly over the charge densities ψ_i in $\mathbb{R}^3$, and the primitive ontology carries the calculation rather than glossing it. The theory is moreover *applied*: the same functional (3) returns atomic energies across the periodic table, molecular geometries and bond energies, reaction paths, condensed phases, and — in a proton–electron reading — nuclear binding. These are not paper derivations but a large public library of interactive, browser-runnable simulations, collected in the RealQM gallery [12], where the charge densities of atoms, molecules, reactions, and nuclei can be computed and watched directly. The metaphysical significance is that RealQM’s ontology is not posited for interpretive tidiness but *earns its keep by doing the calculation*: it is a working, applied method of which the ontology of §§2–3 is the direct reading. An ontology that is also the algorithm is a stronger kind of proposal than one available only as an interpretation.

5 The point and the field: a perspective

The extension at issue reaches past non-relativistic quantum mechanics. The Standard Model’s Lagrangian carries three gauge forces and a Higgs and Yukawa sector,

$$\mathcal{L}_{\text{SM}} = -\frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu} + \bar{\psi} i\gamma^\mu D_\mu \psi + |D_\mu \Phi|^2 - V(\Phi) - (y \bar{\psi}_L \Phi \psi_R + \text{h.c.}), \quad (5)$$

its fundamental carriers *point* excitations of quantum fields. That is a second departure from *res extensa*: where (1) moved the object into $\mathbb{R}^{3N}$, field theory shrank it to a point in $\mathbb{R}^3$. The point carries the classical infinity of a zero-extension charge’s self-energy, softened but not cured by renormalisation — and it is telling that string theory restores extension (to strings) precisely to tame it. Retaining only the electromagnetic term of $\mathcal{L}_{\text{SM}}$ and reading it, non-relativistically and electrostatically, as extended densities returns exactly the energy functional (3): RealQM is that one term, read as *res extensa* in $\mathbb{R}^3$. The ontological choice — the point or the extended thing —

is thus one question asked at every scale, and RealQM answers it, in its domain, on the side of extension. Nothing here disputes the Standard Model’s account of the sub-atomic world; the scales differ.

6 Conclusion

The arena of the wave function and the individuality of particles are two readings of the two features of one formula, (1): its $3N$ -dimensional domain and its antisymmetry. RealQM, the real-space charge-density formulation (2)–(3), declines both, and so meets both problems with one move — a primitive ontology in three-space whose electrons are individuated absolutely, by the regions they occupy, because $\Omega_i \cap \Omega_j = \emptyset$ does double duty. Extension and individuation are the same fact, and Pauli exclusion and the identity of indiscernibles coincide. Whether RealQM is empirically adequate is for physics to decide; if it is, its metaphysics is the cleaner one.

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